

13.13 Solution: We cannot simply set

$$\exp \left[i \left(\frac{\omega}{v} - k \cos \theta \right) Z \right]$$

to zero in the expression for $\mathbf{F}$ after Eq. (13.80). Instead, we should use

$$\mathbf{F}' = \left(1 - \exp \left[i \left(\frac{\omega}{v} - k \cos \theta \right) a \right] \right) \mathbf{F} = \exp \left[\frac{ia}{2} \left(\frac{\omega}{v} - k \cos \theta \right) \right] \times \left(-2i \sin \left(\frac{a}{2} \left(\frac{\omega}{v} - k \cos \theta \right) \right) \right) \mathbf{F}.$$

Therefore, the result for a dielectric foil differs from that of a single-interface by a factor of

$$\mathcal{F} = 4 \sin^2 \Theta,$$

where

$$\Theta = \frac{a}{2} \left(\frac{\omega}{v} - k \cos \theta \right).$$

We now need to show

$$\Theta = \nu \left(1 + \frac{1}{\nu^2} + \eta \right) \frac{a}{4D},$$

under the extreme condition $\omega \gg \omega_p$, $\gamma \gg 1$ and $\theta \ll 1$. To this end, we have

$$\begin{aligned} \frac{\omega}{v} - k \cos \theta &= \frac{\omega}{v} - \frac{\omega}{c} \sqrt{\epsilon(\omega)} \cos \theta = \frac{\omega}{c} \left(\frac{1}{\beta} - \sqrt{\epsilon(\omega)} \cos \theta \right) \\ &= \frac{\omega}{c} \left[\left(1 - \frac{1}{\gamma^2} \right)^{-1/2} - \left(1 - \frac{\omega_p^2}{\omega^2} \right)^{1/2} \left(1 - \frac{\theta^2}{2} \right) \right] \\ &= \frac{\omega}{c} \left[1 + \frac{1}{2\gamma^2} - 1 + \frac{\omega_p^2}{2\omega^2} + \frac{\theta^2}{2} \right] \\ &= \frac{\omega}{c} \left[\frac{1}{2\gamma^2} + \frac{\omega_p^2}{2\omega^2} + \frac{\theta^2}{2} \right] \\ &= \frac{\omega}{2\gamma^2 c} \left(1 + \frac{\gamma^2 \omega_p^2}{\omega^2} + \gamma^2 \theta^2 \right) = \frac{\omega}{2\gamma^2 c} \left(1 + \frac{1}{\nu^2} + \eta \right) \\ &= \frac{\omega_p}{2\gamma c} \frac{\omega}{\gamma \omega_p} \left(1 + \frac{1}{\nu^2} + \eta \right) \\ &= \frac{\nu}{2D} \left(1 + \frac{1}{\nu^2} + \eta \right), \end{aligned}$$

and

$$\Theta = \frac{a}{2} \left(\frac{\omega}{v} - k \cos \theta \right) = \nu \left(1 + \frac{1}{\nu^2} + \eta \right) \frac{a}{2D}.$$